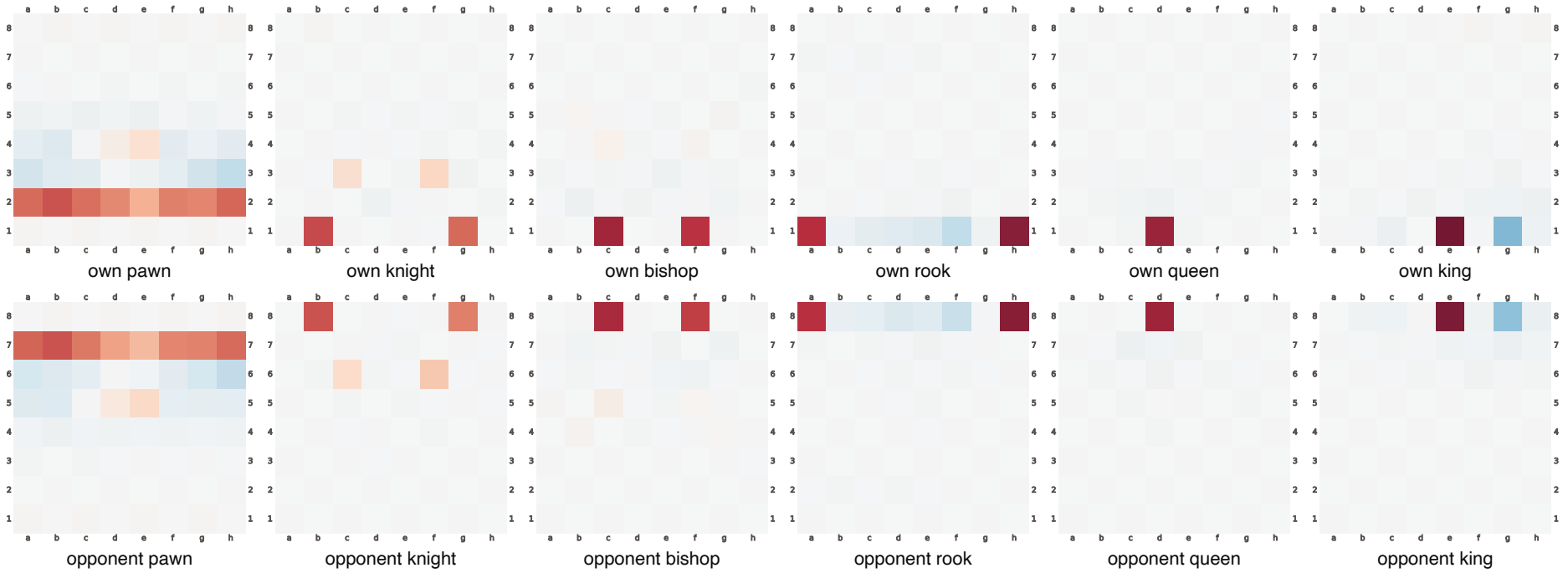
Maia-2 failures by time pressure

Time pressure: more than 1 min

more than 1 min: 583688 non-forced positions | failure rate 0.508 | phase mix {'middlegame': 0.43, 'opening': 0.41, 'endgame': 0.16}

PC1 loadings, more than 1 min (16.6% of variance), side to move at the bottom

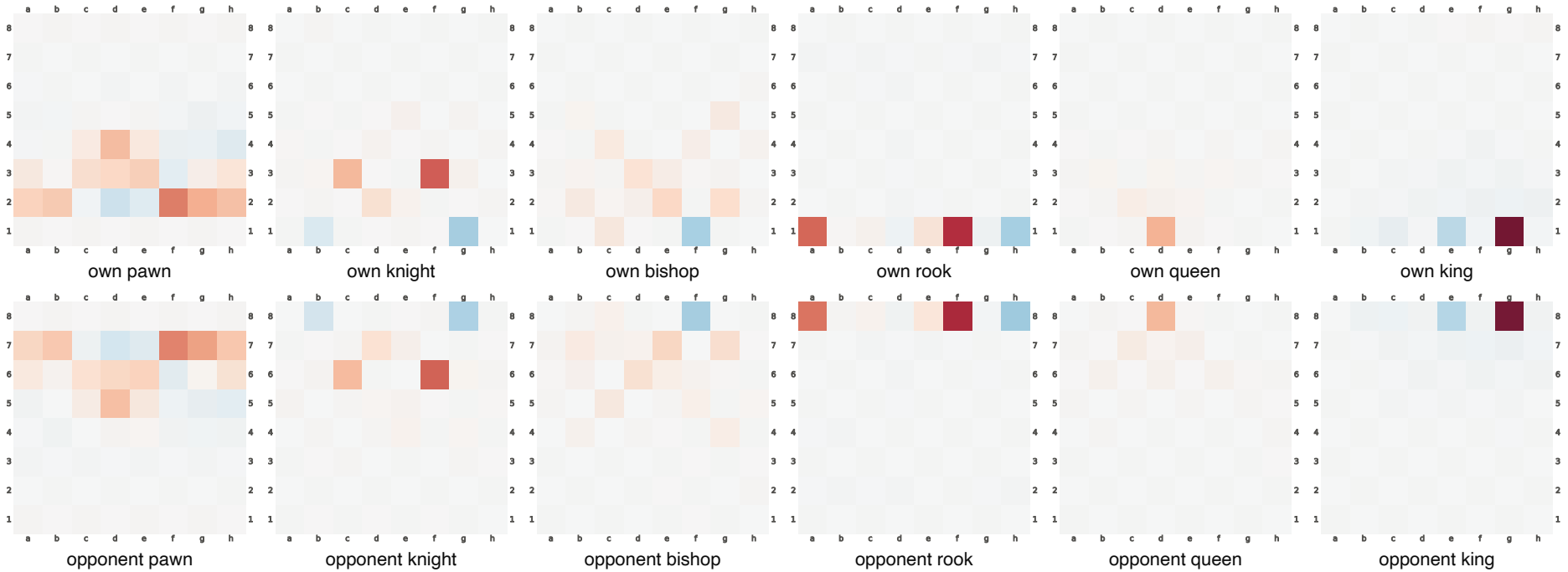


red = a piece on this square raises the PC score, blue = lowers it

-0.241 0.241

more than 1 min PC1 failure rate by score decile (low -> high): [0.45, 0.462, 0.486, 0.502, 0.458, 0.506, 0.503, 0.508, 0.518, 0.736]

PC2 loadings, more than 1 min (4.3% of variance), side to move at the bottom

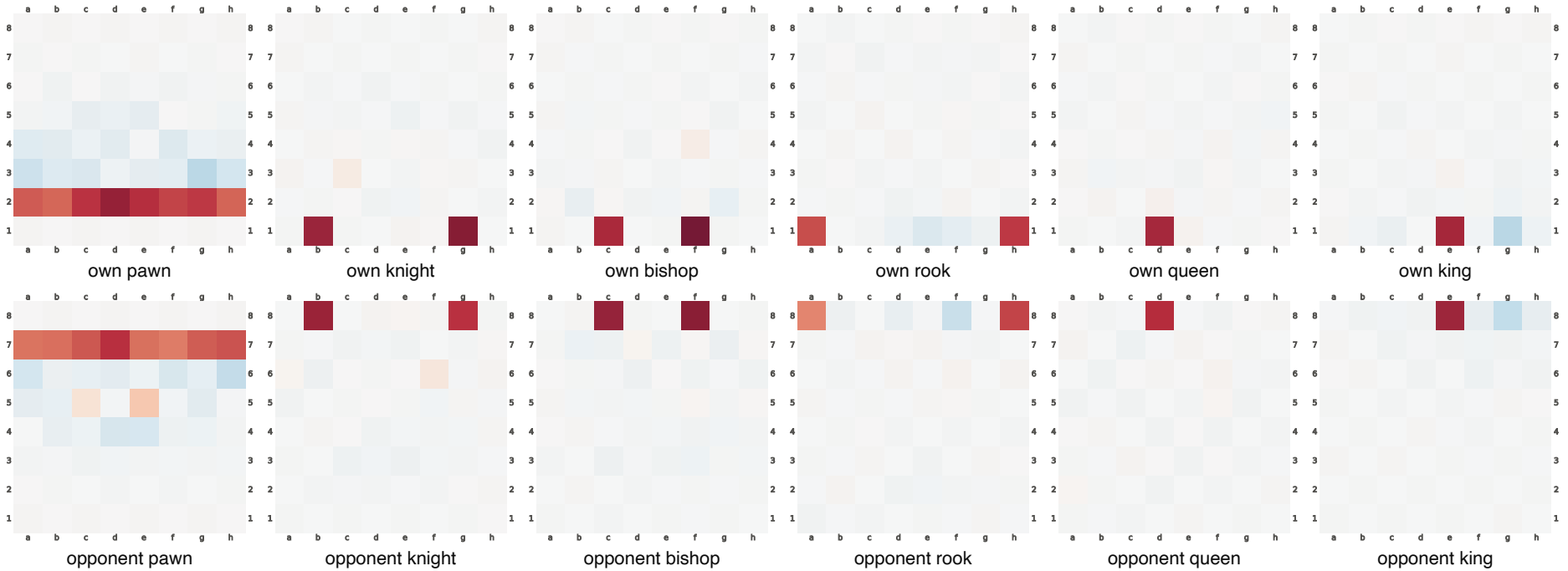


red = a piece on this square raises the PC score, blue = lowers it

-0.335 0.335

more than 1 min PC2 failure rate by score decile (low -> high): [0.449, 0.644, 0.514, 0.503, 0.503, 0.516, 0.477, 0.5, 0.51, 0.514]

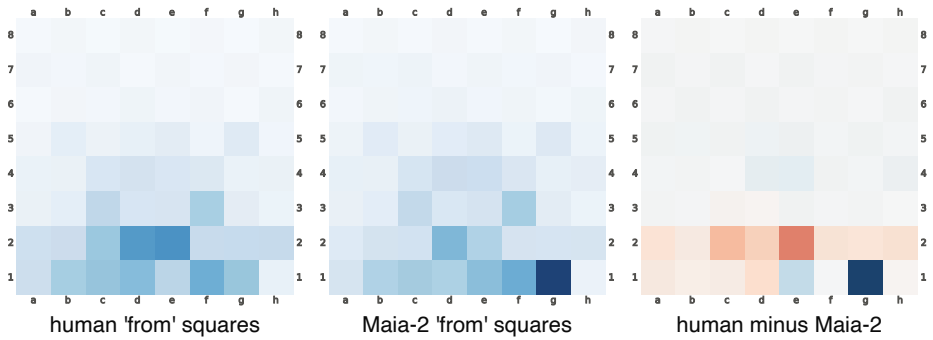
Where pieces stand when Maia-2 fails vs. succeeds (more than 1 min, n=20000)



percentage points: red = occupied more often in failed positions, blue = in correct ones

-9.27 9.27

Failed predictions, more than 1 min: 'from' squares (n=296514, mover at the bottom)



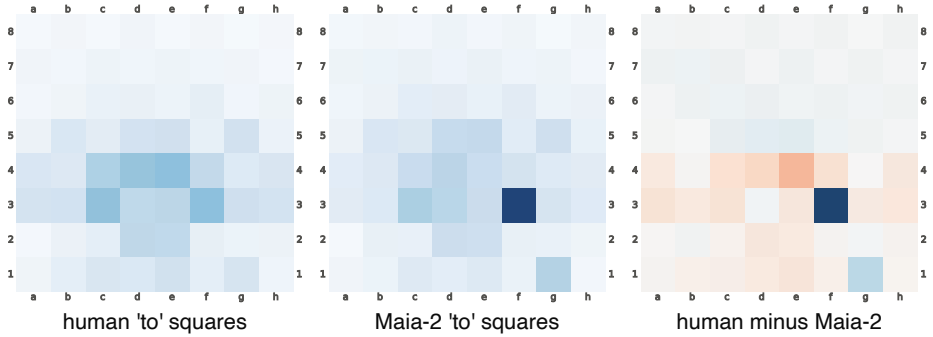
share of moves in %

0 11.2

red = humans use this square more, blue = Maia-2 expects it more (pp)

-6.67 6.67

Failed predictions, more than 1 min: 'to' squares (n=296514, mover at the bottom)



human 'to' squares | Maia-2 'to' squares | human minus Maia-2

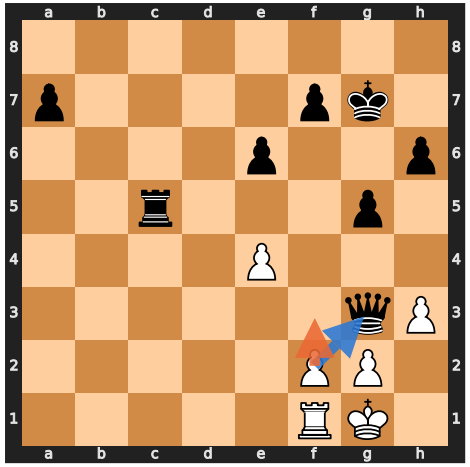
share of moves in %

0 — 12.5

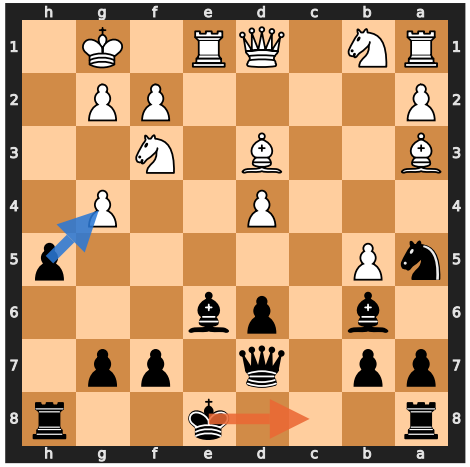
red = humans use this square more, blue = Maia-2 expects it more (pp)

-6.87 — 6.87

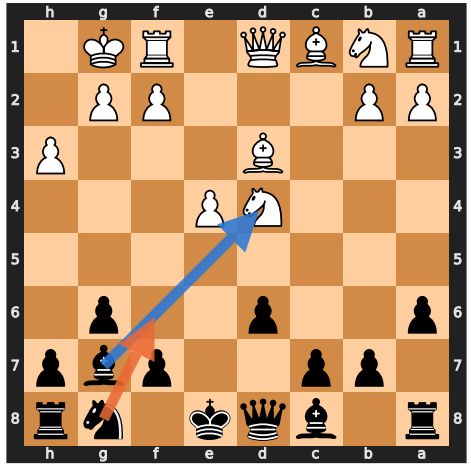
Most confident Maia-2 failures (more than 1 min)



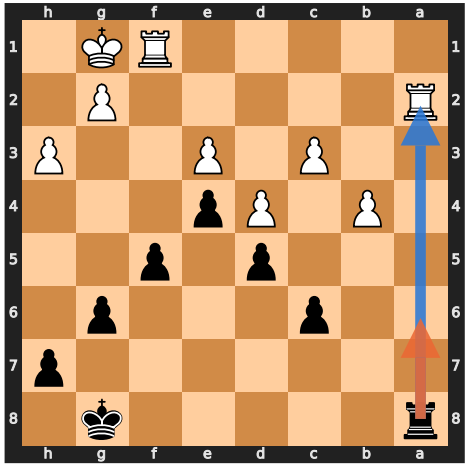
white to move, 2047 vs 2079, 183s left
Maia-2 f2g3 (100%) | human f2f3 (0%)
Pirc Defense



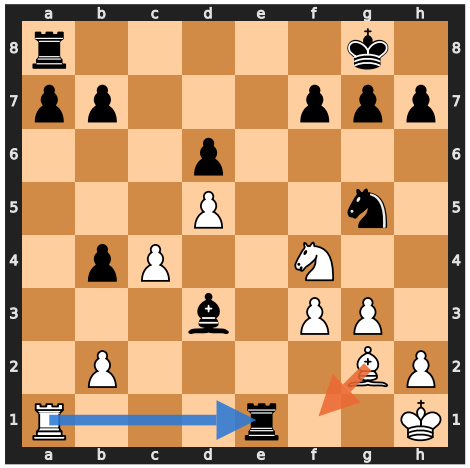
black to move, 2096 vs 2160, 109s left
Maia-2 h5g4 (100%) | human e8c8 (0%)
Italian Game: Classical Variation, Center Attack



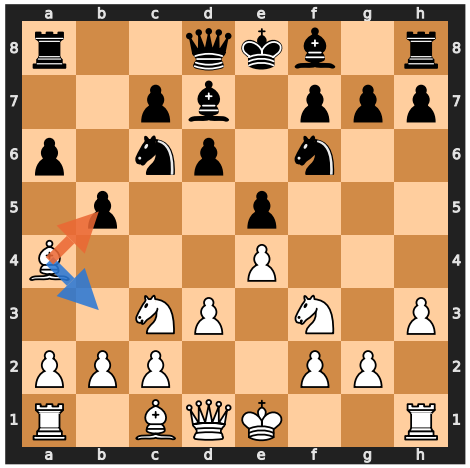
black to move, 2022 vs 1919, 707s left
Maia-2 g7d4 (100%) | human g8f6 (0%)
Pirc Defense



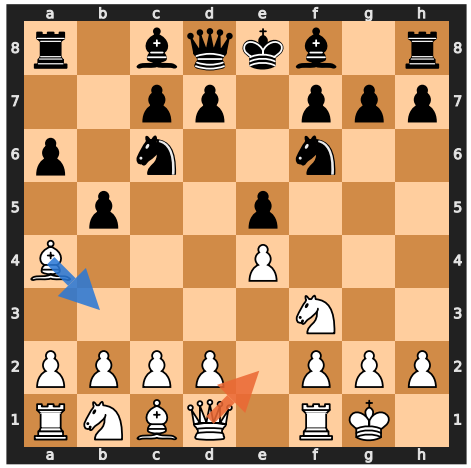
black to move, 1053 vs 1013, 404s left
Maia-2 a8a2 (100%) | human a8a6 (0%)
Queen's Pawn Game: Accelerated London System



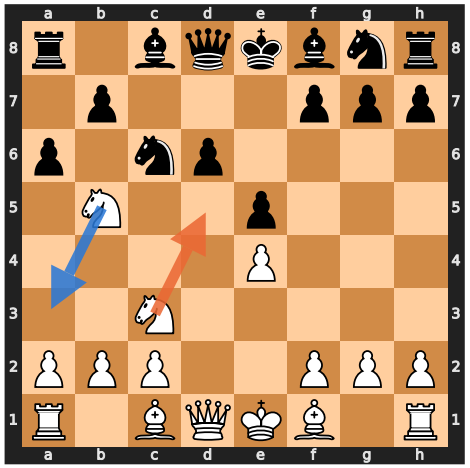
white to move, 1375 vs 941, 193s left
Maia-2 a1e1 (100%) | human g2f1 (0%)
English Opening: King's English Variation, Reversed Closed Sicilian



white to move, 1646 vs 1678, 586s left
Maia-2 a4b3 (100%) | human a4b5 (0%)
Four Knights Game: Spanish Variation



white to move, 2078 vs 2083, 177s left
Maia-2 a4b3 (100%) | human d1e2 (0%)
Ruy Lopez: Morphy Defense



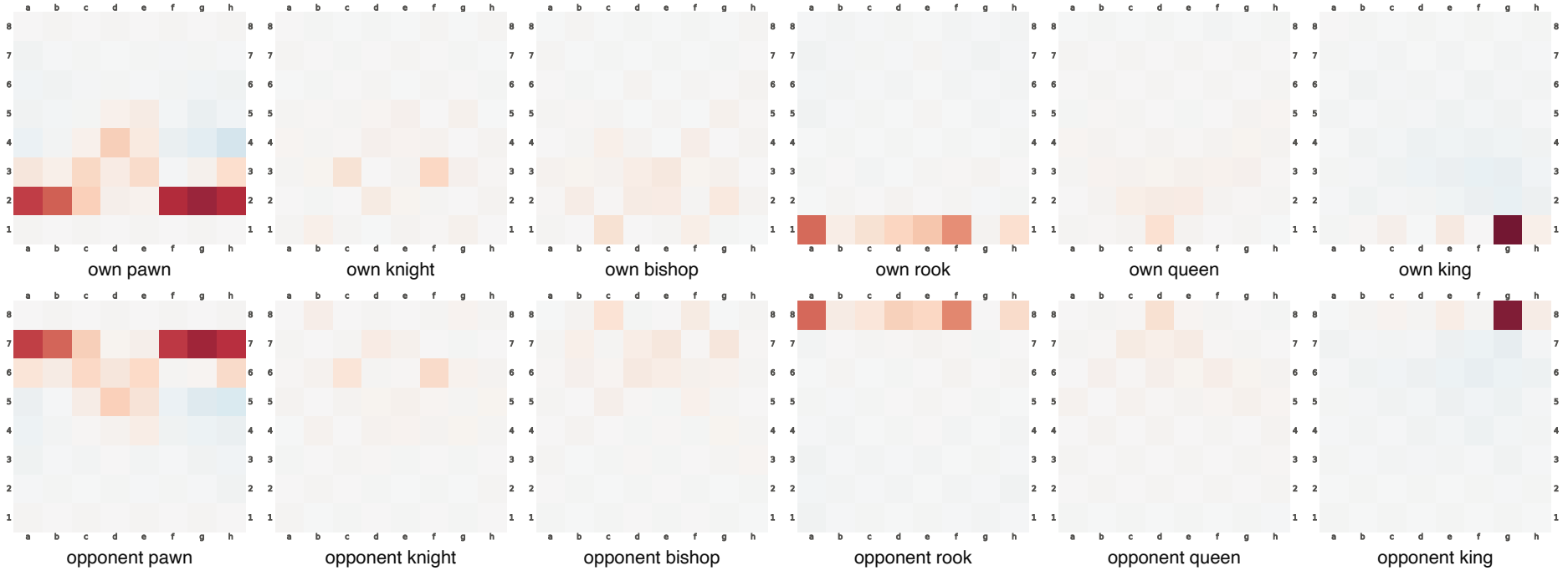
white to move, 2565 vs 2569, 175s left
Maia-2 b5a3 (100%) | human c3d5 (0%)
Sicilian Defense: Kalashnikov Variation

blue arrow = Maia-2 top prediction, orange arrow = move actually played

Time pressure: 30 s to 1 min

30 s to 1 min: 38371 non-forced positions | failure rate 0.459 | phase mix {'endgame': 0.57, 'middlegame': 0.43, 'opening': 0.01}

PC1 loadings, 30 s to 1 min (5.1% of variance), side to move at the bottom

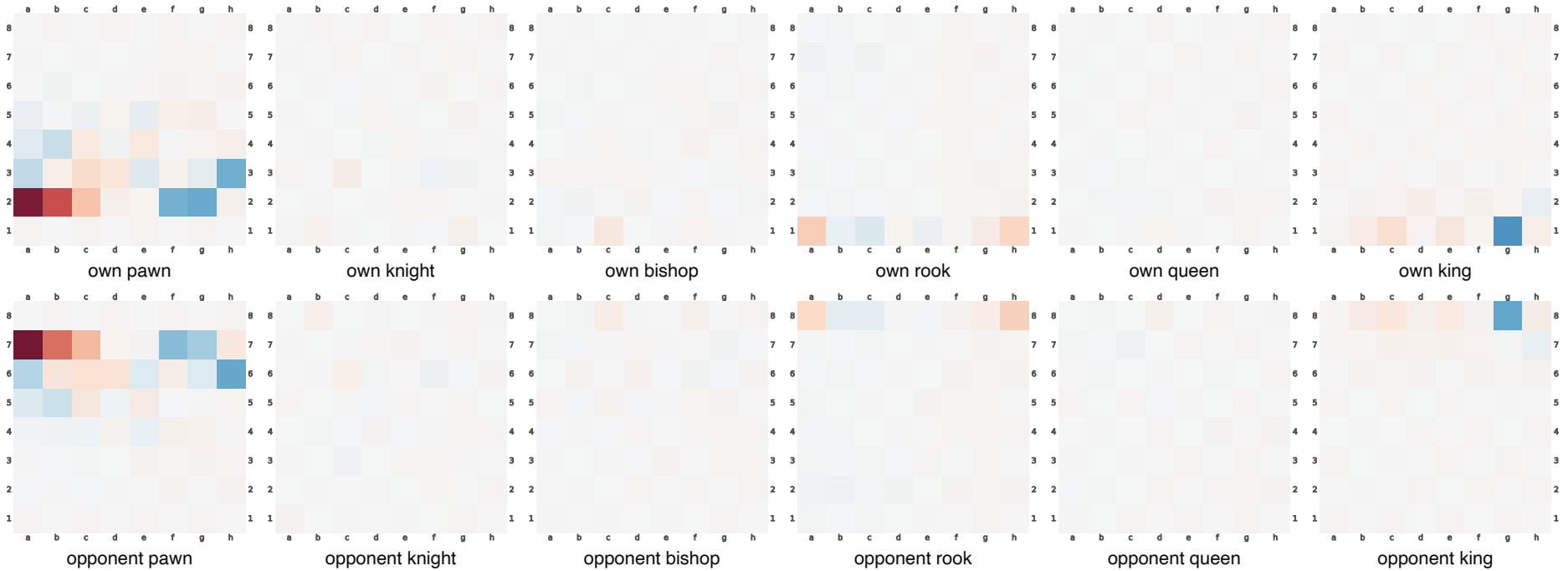


red = a piece on this square raises the PC score, blue = lowers it

-0.304 0.304

30 s to 1 min PC1 failure rate by score decile (low -> high): [0.397, 0.436, 0.436, 0.458, 0.452, 0.458, 0.462, 0.486, 0.496, 0.49]

PC2 loadings, 30 s to 1 min (2.6% of variance), side to move at the bottom

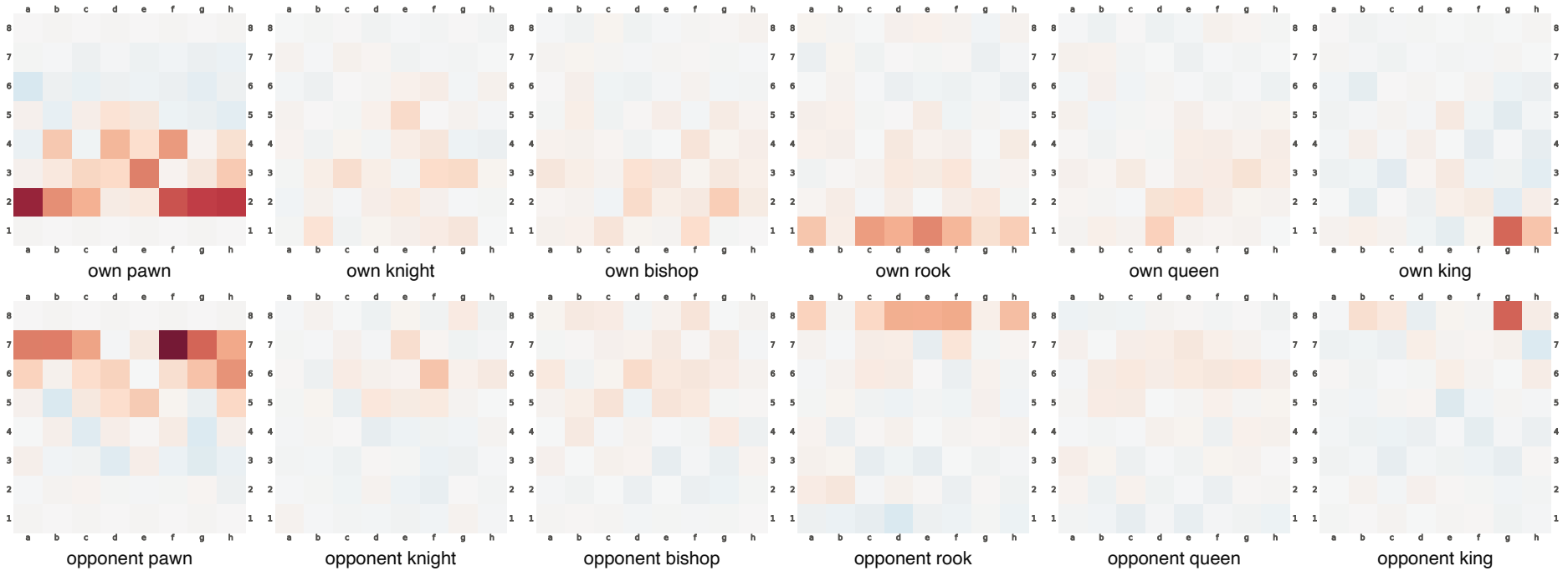


red = a piece on this square raises the PC score, blue = lowers it

-0.4 0.4

`30 s to 1 min PC2 failure rate by score decile (low -> high): [0.481, 0.456, 0.448, 0.462, 0.442, 0.416, 0.457, 0.47, 0.468, 0.474]`

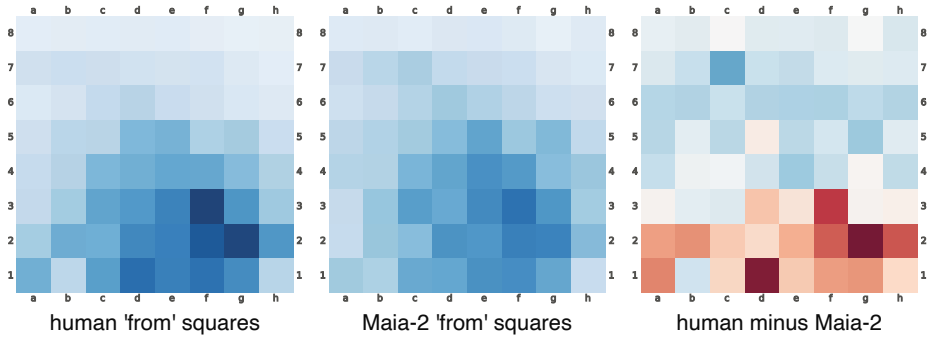
Where pieces stand when Maia-2 fails vs. succeeds (30 s to 1 min, n=20000)



percentage points: red = occupied more often in failed positions, blue = in correct ones

-3.19 3.19

Failed predictions, 30 s to 1 min: 'from' squares (n=17594, mover at the bottom)



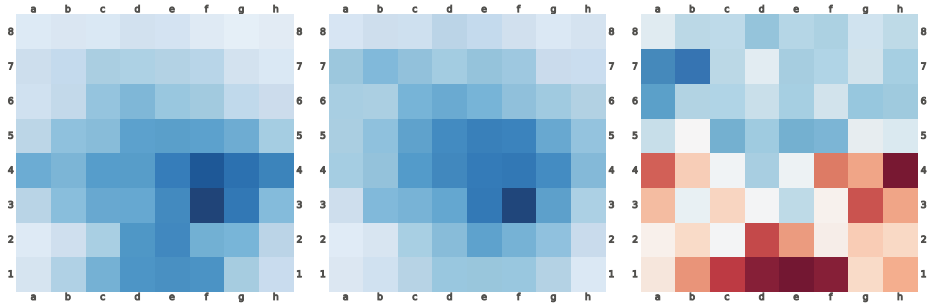
share of moves in %

0 4

red = humans use this square more, blue = Maia-2 expects it more (pp)

-1.03 1.03

Failed predictions, 30 s to 1 min: 'to' squares (n=17594, mover at the bottom)



human 'to' squares | Maia-2 'to' squares | human minus Maia-2

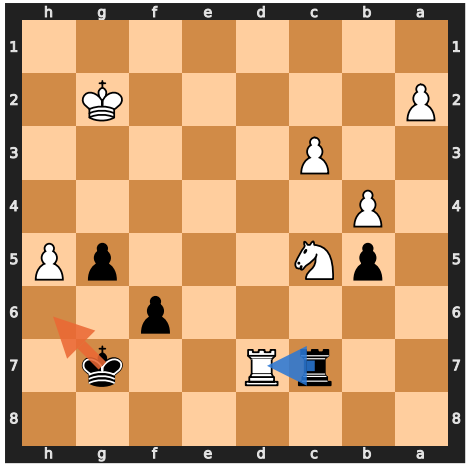
share of moves in %

0 3.68

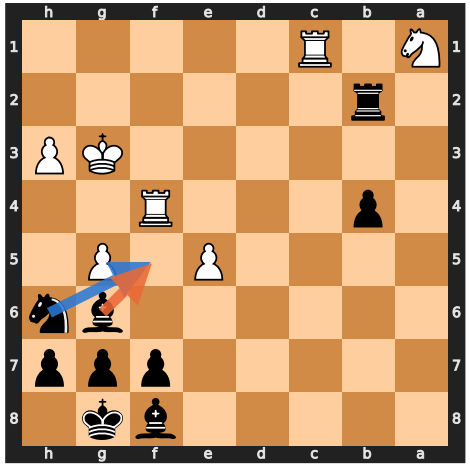
red = humans use this square more, blue = Maia-2 expects it more (pp)

-0.955 0.955

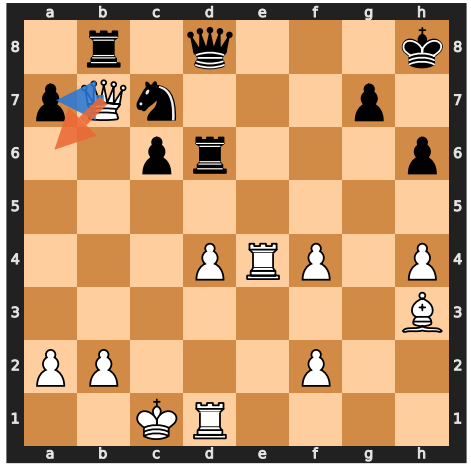
Most confident Maia-2 failures (30 s to 1 min)



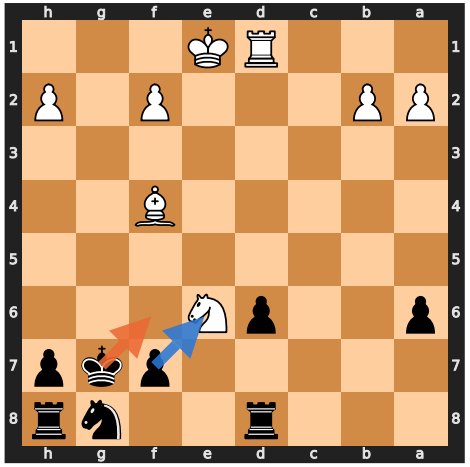
black to move, 1860 vs 1854, 50s left
Maia-2 c7d7 (100%) | human g7h6 (0%)
Caro-Kann Defense



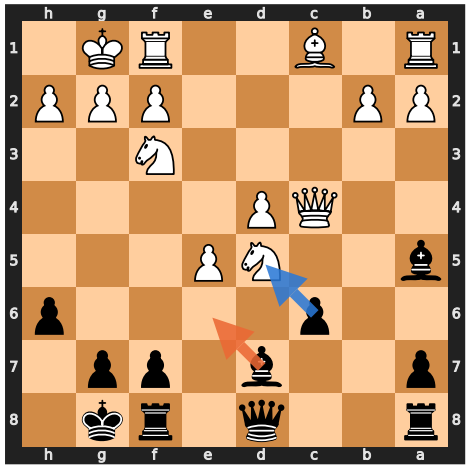
black to move, 2212 vs 2255, 54s left
Maia-2 h6f5 (100%) | human g6f5 (0%)
Sicilian Defense: McDonnell Attack



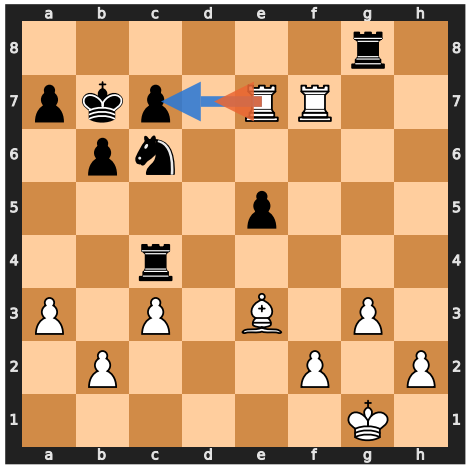
white to move, 1814 vs 2171, 37s left
Maia-2 b7a7 (100%) | human b7a6 (0%)
English Opening: Anglo-Dutch Defense



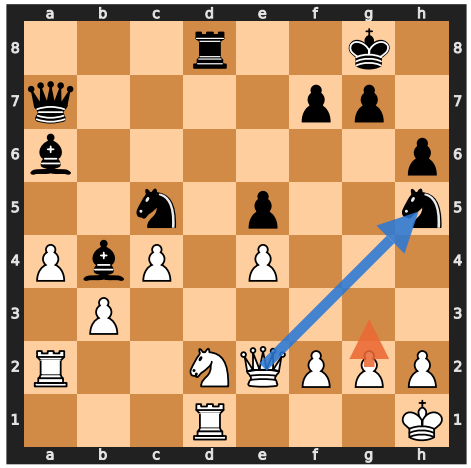
black to move, 1600 vs 1673, 51s left
Maia-2 f7e6 (100%) | human g7f6 (0%)
Scotch Game



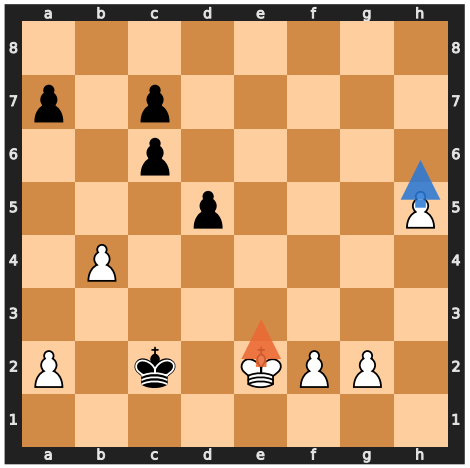
black to move, 1464 vs 1465, 47s left
Maia-2 c6d5 (99%) | human d7e6 (0%)
Italian Game: Two Knights Defense, Fritz Variation



white to move, 1857 vs 1827, 46s left
Maia-2 e7c7 (99%) | human e7d7 (0%)
Italian Game: Rousseau Gambit



white to move, 1866 vs 1604, 39s left
Maia-2 e2h5 (99%) | human g2g3 (0%)
Philidor Defense: Hanham Variation



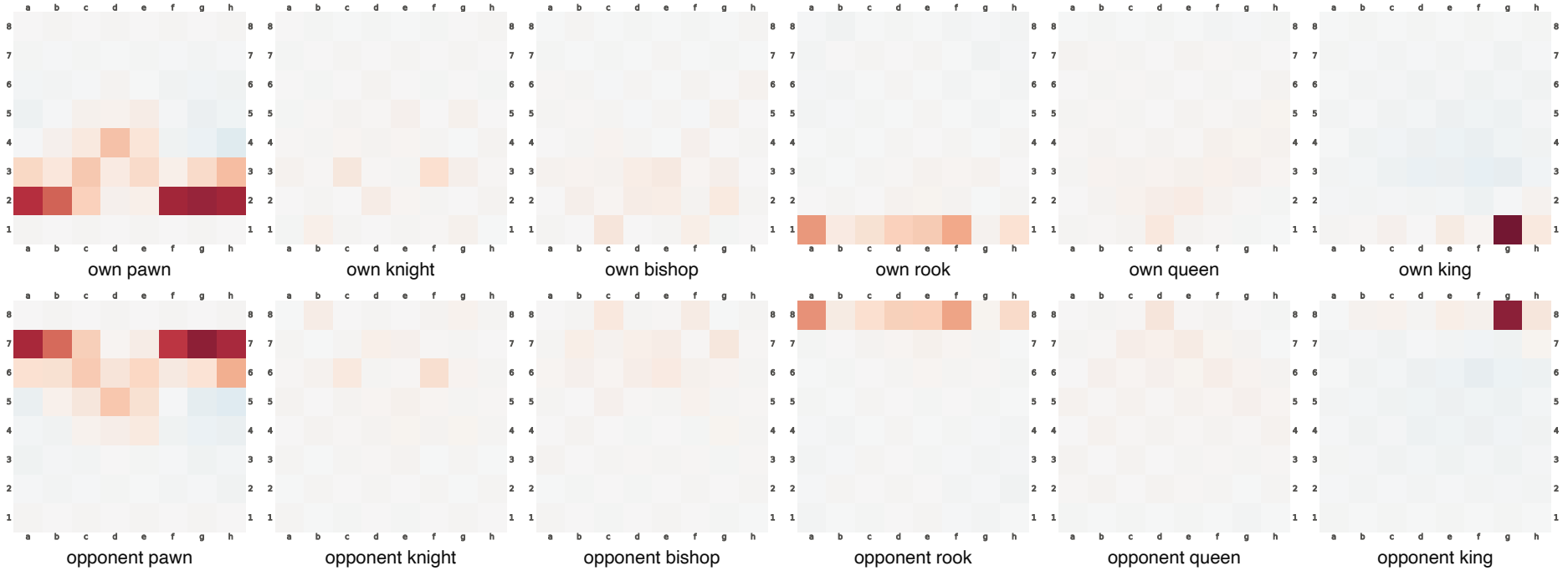
white to move, 2173 vs 1950, 52s left
Maia-2 h5h6 (99%) | human e2e3 (0%)
Italian Game: Two Knights Defense, Open Variation

blue arrow = Maia-2 top prediction, orange arrow = move actually played

Time pressure: less than 30 s

less than 30 s: 47700 non-forced positions | failure rate 0.468 | phase mix {'endgame': 0.71, 'middlegame': 0.28, 'opening': 0.0}

PC1 loadings, less than 30 s (4.7% of variance), side to move at the bottom

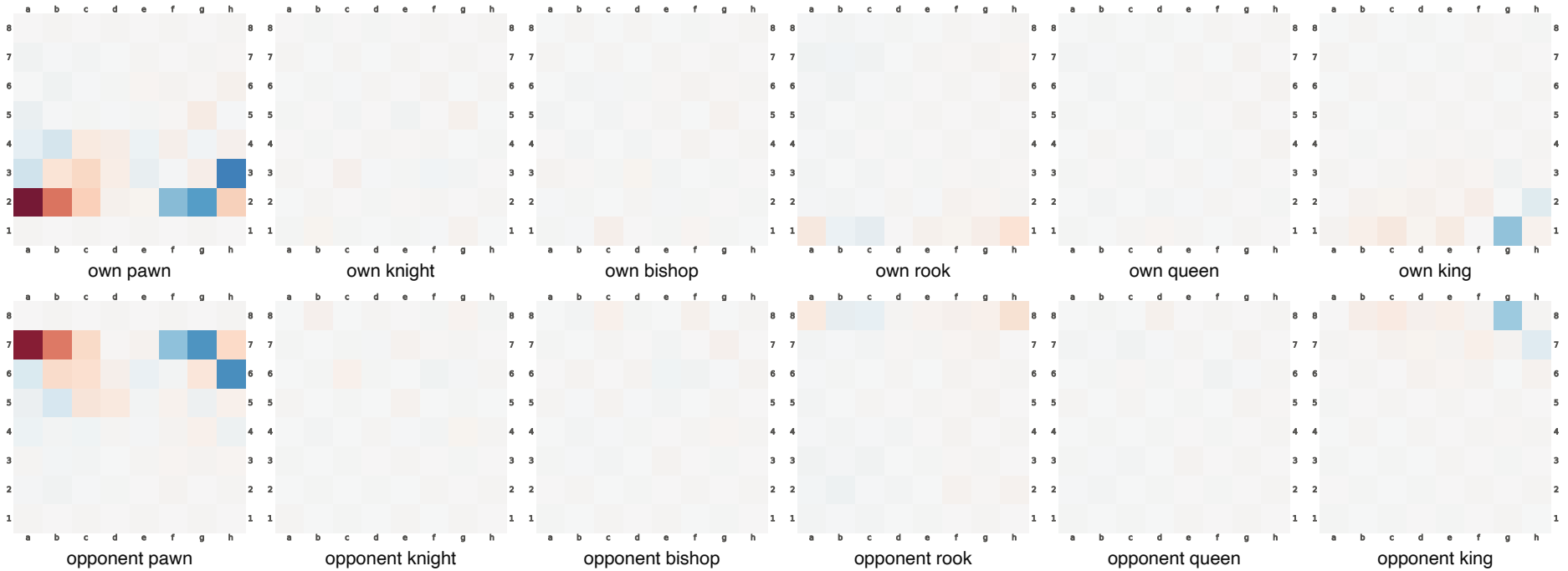


red = a piece on this square raises the PC score, blue = lowers it

-0.299 0.299

less than 30 s PC1 failure rate by score decile (low -> high): [0.467, 0.442, 0.456, 0.447, 0.454, 0.472, 0.486, 0.458, 0.485, 0.492]

PC2 loadings, less than 30 s (2.4% of variance), side to move at the bottom

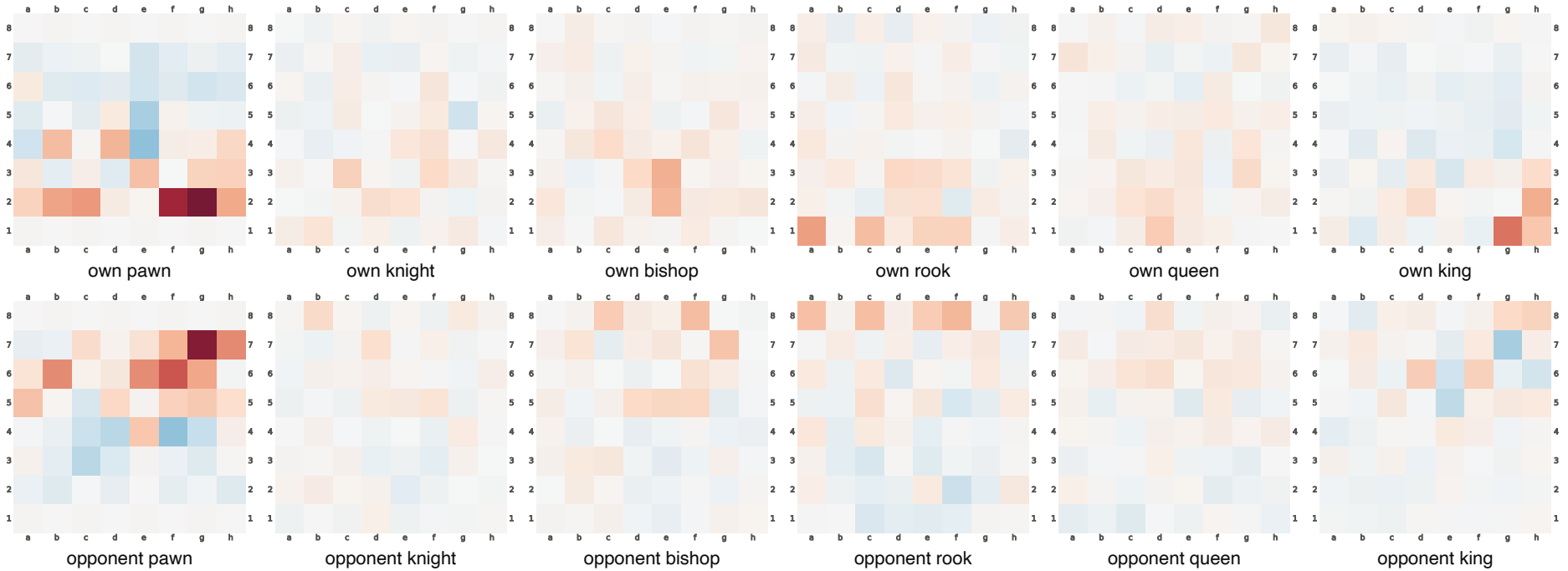


red = a piece on this square raises the PC score, blue = lowers it

-0.415 0.415

less than 30 s PC2 failure rate by score decile (low -> high): [0.476, 0.498, 0.466, 0.45, 0.441, 0.486, 0.471, 0.46, 0.438, 0.474]

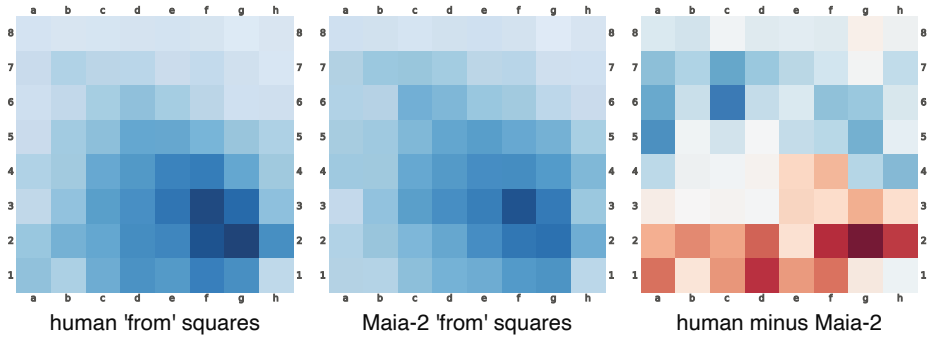
Where pieces stand when Maia-2 fails vs. succeeds (less than 30 s, n=20000)



percentage points: red = occupied more often in failed positions, blue = in correct ones

-1.98 1.98

Failed predictions, less than 30 s: 'from' squares (n=22329, mover at the bottom)



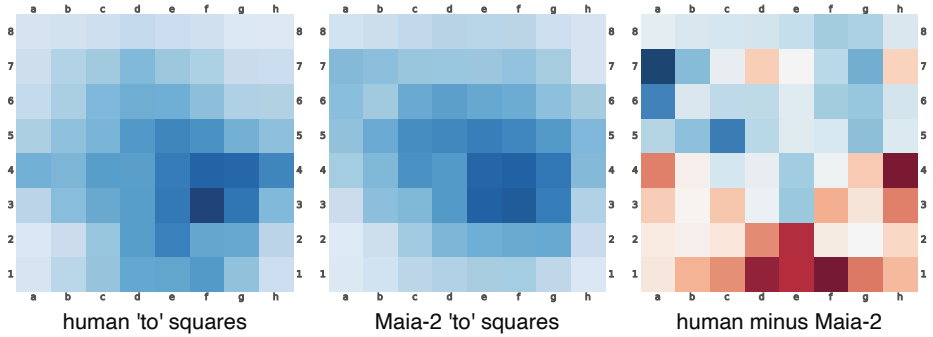
share of moves in %

0 3.52

red = humans use this square more, blue = Maia-2 expects it more (pp)

-0.681 0.681

Failed predictions, less than 30 s: 'to' squares (n=22329, mover at the bottom)



human 'to' squares | Maia-2 'to' squares | human minus Maia-2

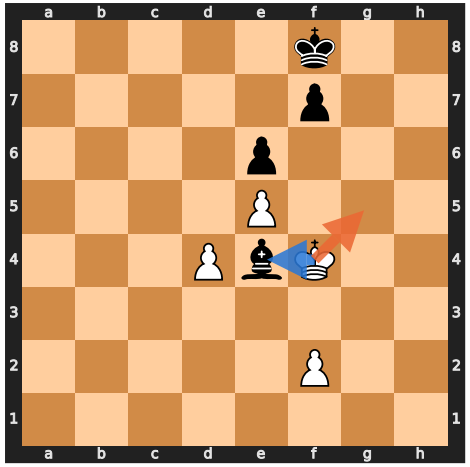
share of moves in %

0 3.45

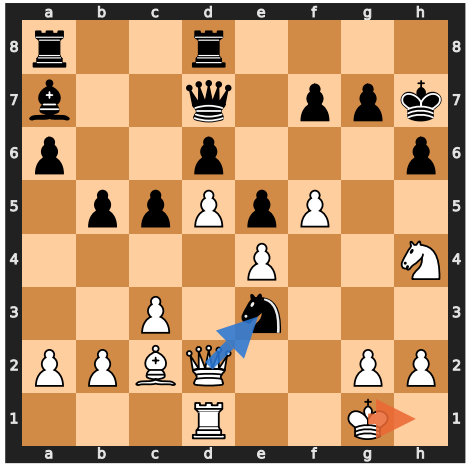
red = humans use this square more, blue = Maia-2 expects it more (pp)

-0.869 0.869

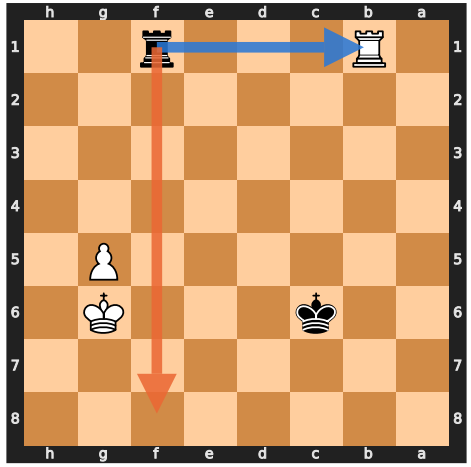
Most confident Maia-2 failures (less than 30 s)



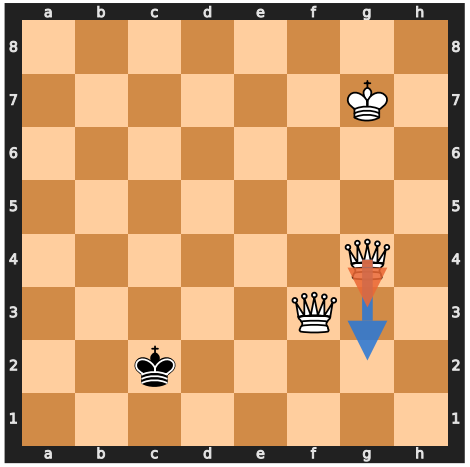
white to move, 2442 vs 2563, 4s left
Maia-2 f4e4 (100%) | human f4g5 (0%)
Grünfeld Defense: Exchange Variation, Spassky Variation



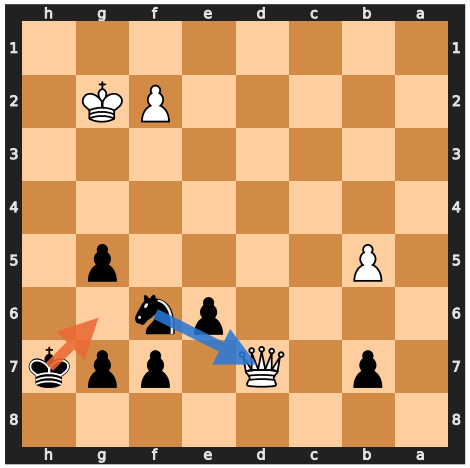
white to move, 1893 vs 1880, 7s left
Maia-2 d2e3 (100%) | human g1h1 (0%)
Ruy Lopez: Morphy Defense, Neo-Arkhangelsk Variation



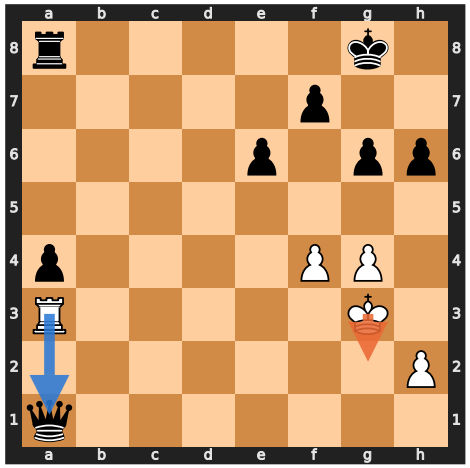
black to move, 1565 vs 1563, 2s left
Maia-2 f1b1 (100%) | human f1f8 (0%)
Caro-Kann Defense: Hillbilly Attack



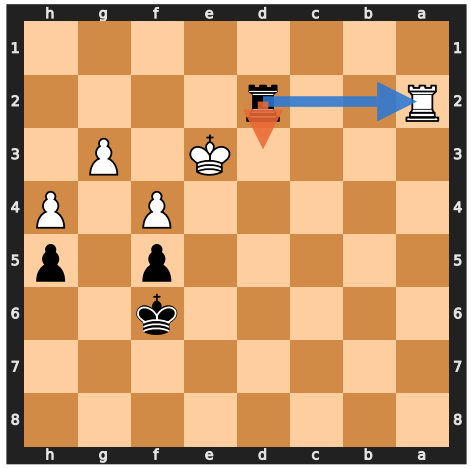
white to move, 1519 vs 1527, 1s left
Maia-2 g4g2 (100%) | human g4g3 (0%)
Sicilian Defense: Old Sicilian



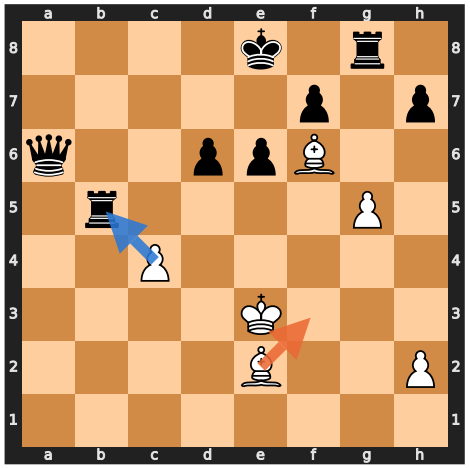
black to move, 2118 vs 2136, 4s left
Maia-2 f6d7 (100%) | human h7g6 (0%)
Caro-Kann Defense: Exchange Variation



white to move, 2482 vs 2461, 5s left
Maia-2 a3a1 (100%) | human g3g2 (0%)
Sicilian Defense: Alapin Variation



black to move, 1909 vs 1964, 13s left
Maia-2 d2a2 (100%) | human d2d3 (0%)
English Opening: Symmetrical Variation, Ultra-Symmetrical Variation



white to move, 2239 vs 2304, 10s left
Maia-2 c4b5 (100%) | human e2f3 (0%)
Owen Defense

blue arrow = Maia-2 top prediction, orange arrow = move actually played